



AUT@MIATA

Technology Services LLC

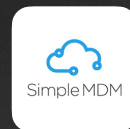
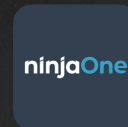
Apple Device Management

Introduction to DMS / MDM

Lecture

Mobile Device Management

- Mobile Device Management (MDM) solutions are available from a variety of vendors including...
 - Mosyle
 - Jamf
 - Kandji
 - Microsoft
 - & many more...



Mobile Device Management

- Key Features of MDM Solutions...
 - Deploy and Manage devices over-the-air.
 - Zero-touch configuration.
 - Configure settings, restrictions, install certificates, etc.
 - Install and update applications or apps.
 - Remote wipe, lock, or erase lost/stolen devices.

Managing Devices

- All MDM/DMS solutions follow a common framework...
 - Enrollment
 - Configuration Profiles
 - Payloads
 - Remote Commands
- Apple Requires an MDM/DMS for Apple Platform Management
- Review the Apple Platform Deployment Guide for API Detail

Apple's MDM Framework

Lecture

Apple's MDM Protocol

- Built into every Apple operating system.
- The protocol is what allows a device to respond to an MDM's command and knows what to do.
- Key functions enabled by the protocol include:
 - Enabling device management, assignment, enrollment.
 - Configuration, commands, queries, status reports.
 - Declarations, check-in, content management.



How MDM Works

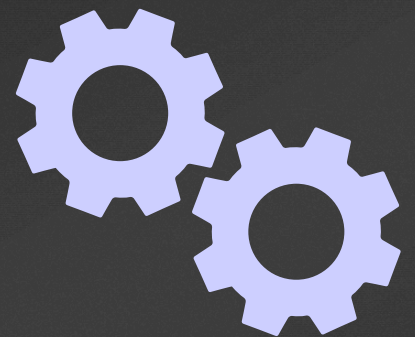
Lecture

How MDM Works

- All MDM/DMS solutions perform four key things...
 - Enrollment
 - Configuration Profiles
 - Payloads
 - Remote Commands

Enrolling Devices

- User Enrollment
 - Account Driven User Enrollment
 - Profile-based User Enrollment (Deprecated)
- Device Enrollment
 - Profile-based Device Enrollment
 - Account Driven Device Enrollment
 - Automated Device Enrollment
 - Supervised Devices vs Not Supervised



Configuration Profiles

- Profiles provide specific settings for one or more configuration.
- Profiles contain one or more payloads.
- The MDM will always install at least one Profile even if it's not managing any other aspect of the device.



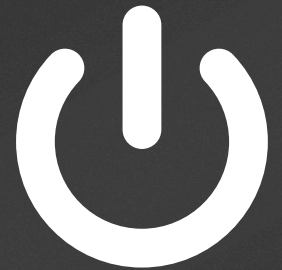
Payloads

- Payloads are the actual configuration settings that a Profile applies to the device.
- Payloads automate the configuration of settings, accounts, restrictions, and more.

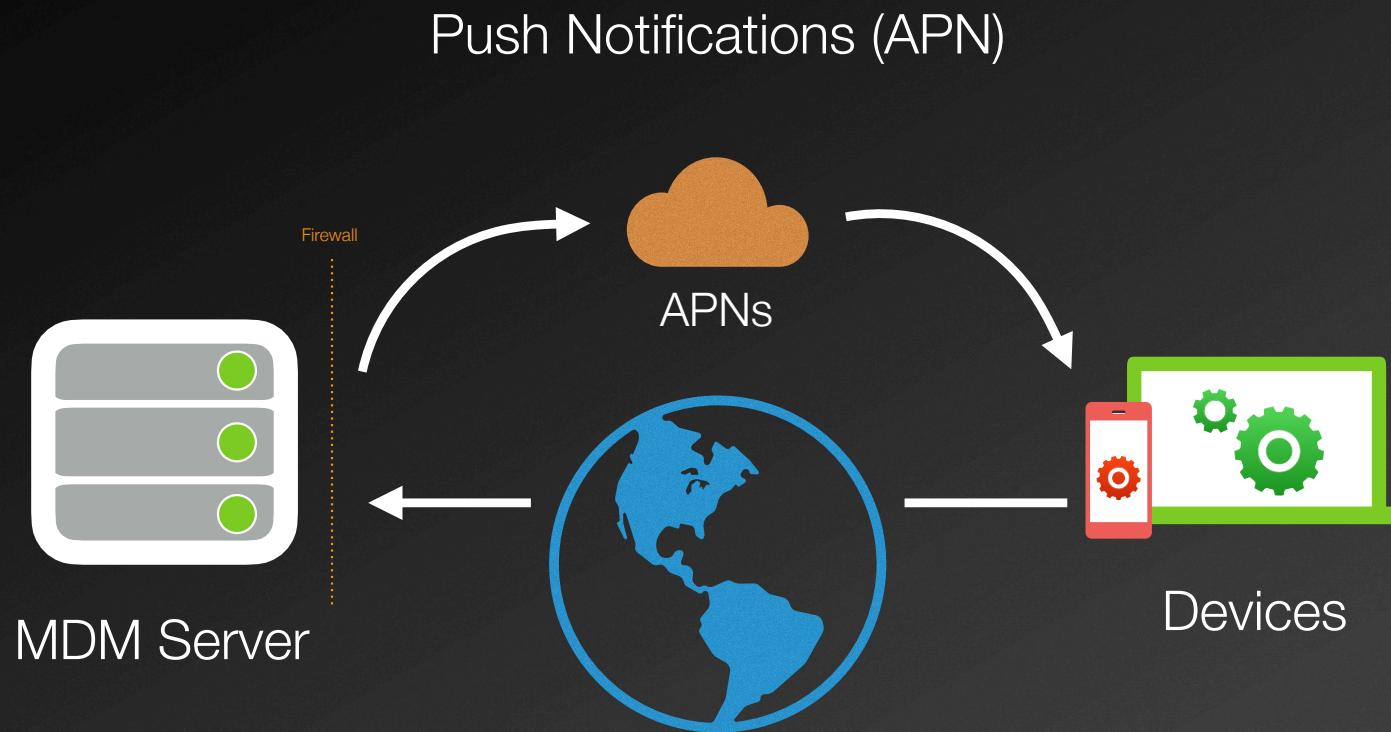


Remote Commands

- Every MDM communicates with its managed devices 'over-the-air' through the Internet.
- MDM Administrators can poll managed devices for information on state.
- MDM Administrators can also send specific commands and scripts to managed devices.
- Declarative Device Management.

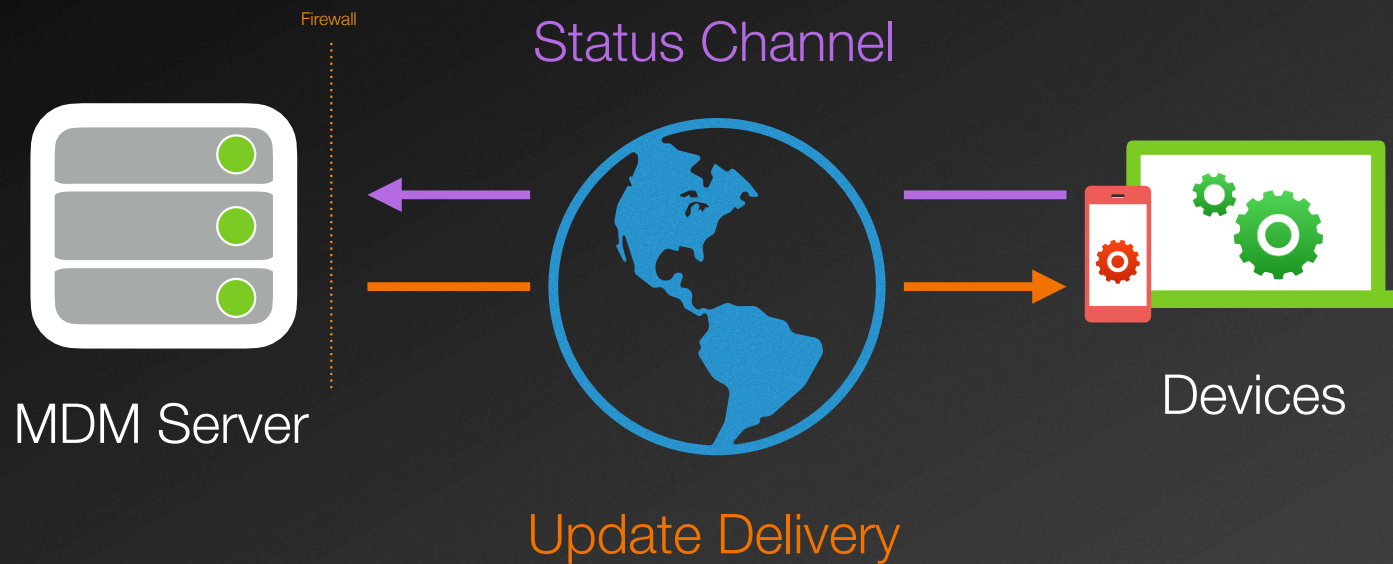


MDM Communication



MDM Communication

Declarative Device Management



Device Enrollment

Lecture

Device Enrollment



Device Enrollment



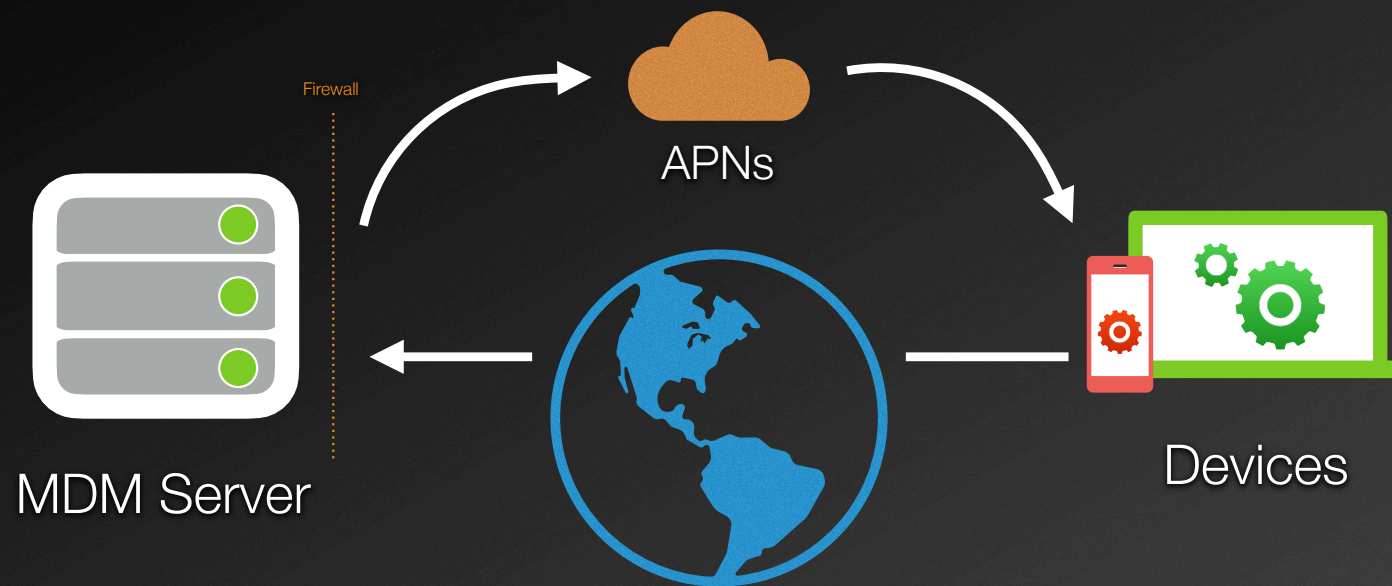
Device Enrollment



Device Enrollment



Communication



MDM Certificates

- MDMs require a number of installed certificates on the device to work correctly.
 - Apple Push Notification certificate
 - Secure Socket Layer (SSL) certificate
- You may see other certificates or profiles on a managed device just needed to establish a management baseline.

Enrollment Models

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Personal Devices

- Account Driven User Enrollment
 - End-users enroll their own device into management via a self-service process using a Managed Apple Account or through a third-party application (Company Portal).
 - Sign In to Work or School Account...

Organization-Owned Devices

- Device Enrollment
 - Using a web portal or third party application to enroll in management using a downloaded enrollment profile.
- Supervision Caveats
 - iOS/iPadOS = Not Supervised
 - macOS = Supervised
- Automated Device Enrollment
 - Enrollment through Setup Assistant, requires device in Apple Business/School Manager
 - Enables 'Zero Touch' Workflows
 - Supervised

Personalization Strategy

- Personally Enabled Devices
 - Devices assigned to a specific user (1:1)
- Non-personalized Devices
 - Multiple users sharing the same device (computer lab, kiosk, etc)
- Shared iPad
 - Multiple users share the device, but each user has a personalized experience and their data is stored in iCloud Drive.

Managed Apple Accounts

Lecture

iOS Account Management



Apple Account



Managed Apple Account

iOS Account Management



Apple Account

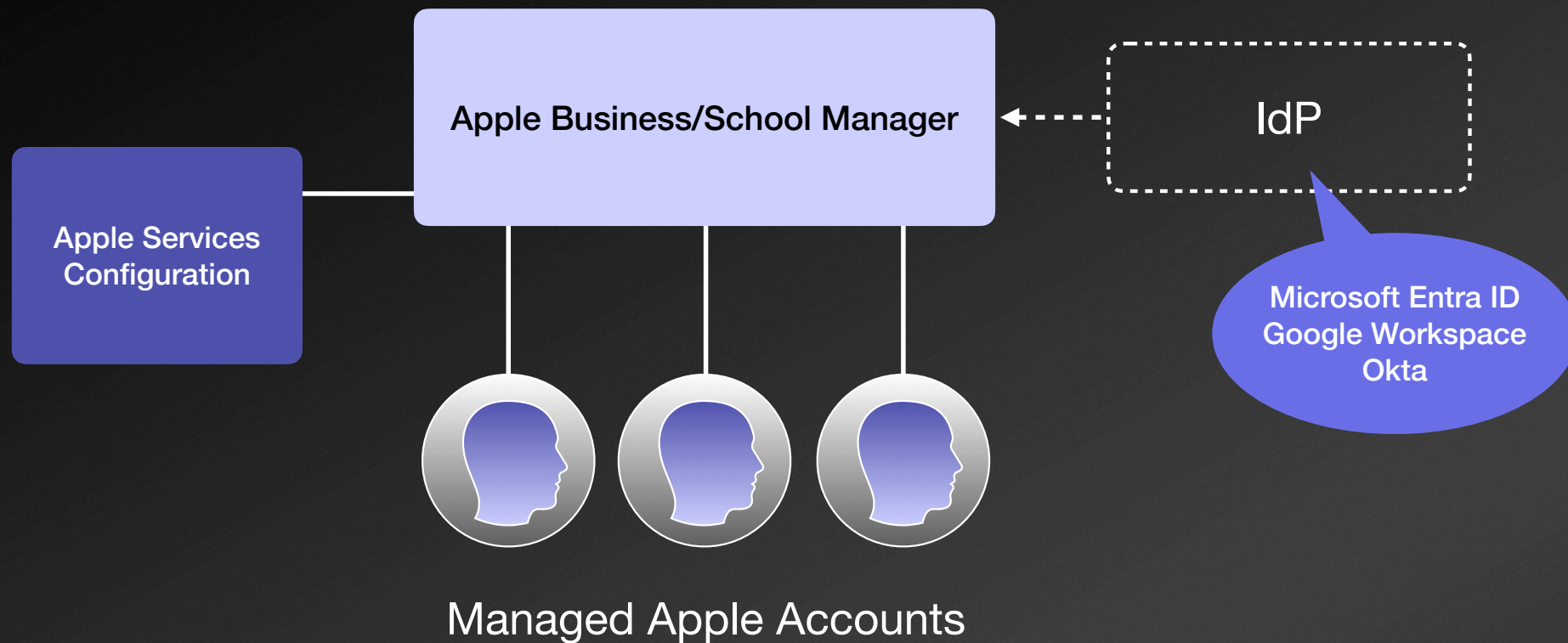
- User creates the Apple Account
- User manages the Apple Account
- User uses the Find My App to locate a lost device
- User has access to create an iCloud Mail account
- Full App Store access & purchasing capability



Managed Apple Account

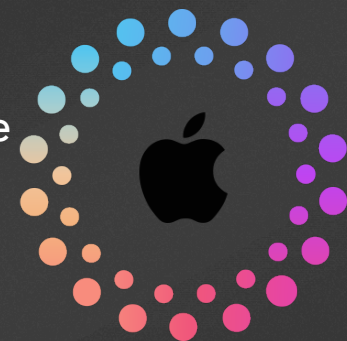
- AXM Administrator creates & assigns
- AXM Administrator managed
- Find My app is disabled, MDM Administrator uses Managed Lost Mode
- No iCloud Mail, organizational email is used instead.
- App Store is browsing only, no ability to purchase

Creating Managed Apple Accounts



Apple Account Considerations

- Personally owned devices (Bring Your Own Device) can be configured with two Apple Accounts...
- A personal Apple Account to allow for things that the Managed Apple Account doesn't provide for or to also sync personal data like your Photo Library.
- A Managed Apple Account enabling organizational options including managed iCloud Drive storage from that organization.
- Organization Owned & Supervised Device settings can enforce only a Managed Apple Account being signed into the device if the device is in Apple Business Manager.



Introduction to AxM

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Introduction to AxM

- Apple School Manager & Apple Business Manager
 - Web-based portals that integrate with your IdP and MDM/DMS.
 - Manage Organizational Settings.
 - Database of Organization-Owned & Purchased Devices.
 - Provisions Managed Apple Accounts.
 - Configure Domain Lock, Federation, and Directory Sync with IdP.
 - Purchase Apps & Books for 'Managed Distribution'.

Sign-Up

 **Business**

Manage your organization's devices, apps,
and accounts.

Apple Account 

☐ Remember me

[Forgot Managed Apple Account or password?](#)
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